

HCM360 HRIS

Data Structure

Schemas, tables, and data flow across the HCM360 HRIS

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1. Database Overview

HCM360 uses PostgreSQL 15 with schema-based partitioning. Every functional module owns its schema to keep concerns isolated. Cross-module references use foreign keys; reporting uses views and materialized views to denormalize for performance.

Primary tenant isolation is through `core.companies.id`. Every employee-scoped table carries either `employee_id` or `company_id` so queries naturally filter by tenant. Row-level security is not enforced; access is filtered at the application layer through role and company context.

2. Core Schema

The core schema holds master data shared across the platform: companies, users, roles, employees, organizational structure, and system configuration.

2.1 Tenant & Company

Table	Key Columns	Notes
core.companies	id, code, name, legal_name, industry, size_bracket, tin, sss_employer_id, phic_employer_id, hdmf_employer_id, fiscal_year_start, payroll_cycle, is_active	One row per subscriber/tenant
core.company_branding	id, company_id, key UNIQUE, value	Key/value settings: short_name, tagline, primary_color, logo_text

2.2 Users & Roles

Table	Key Columns	Notes
core.users	id, company_id, username UNIQUE, email, password_hash, display_name, role_code, employee_id, is_active, last_login_at, failed_attempts, locked_until, mfa_enabled	System login accounts
core.roles	id, code UNIQUE, name, description, is_system_role, is_active	Built-ins: SUPER_ADMIN, HR_ADMIN, MANAGER, EMPLOYEE, EXECUTIVE
core.user_roles	user_id, role_id, granted_by, granted_at	Many-to-many (user can hold multiple roles)

2.3 Employees & Organization

Table	Key Columns
core.employees	id, company_id, employee_no UNIQUE, full_name, gender, civil_status, date_of_birth, mobile_no, work_email, position_id, department_id, employment_type_id, job_grade_id, date_hired, date_regularized, status, immediate_supervisor_id, basic_salary, daily_rate, hourly_rate
core.departments	id, company_id, code, name, parent_id, head_employee_id
core.positions	id, company_id, code, title, department_id, job_grade_id
core.job_grades	id, code, name, grade_level, salary_min, salary_max
core.employment_types	id, code, name, probation_days, is_entitled_benefits
core.business_units	id, company_id, code, name, parent_id

2.4 Employee Sub-Records

Table	Purpose
core.employee_addresses	Permanent / current addresses
core.employee_emergency_contacts	Named contacts with relationship + mobile
core.employee_government_ids	SSS, TIN, PhilHealth, HDMF, TIN, driver license
core.employee_bank_accounts	Bank, account number, primary flag
core.employee_dependents	Spouse, children, beneficiary flag

2.5 Access Control

Table	Purpose
core.page_registry	Every UI route registered with module, nav_group, icon, title
core.role_page_access	Role → page access grant (can_access boolean)
core.user_page_access	User-level override (takes precedence over role)
core.feature_registry	Action-level features (CREATE, EDIT, DELETE, EXPORT, APPROVE)
core.role_feature_access	Role → feature grant
core.user_feature_access	User-level feature override
core.mod_permissions	Module-level write permissions (separate from page access)
core.field_privacy_rules	RA 10173 field visibility rules per (section, field, role)

2.6 Reference & Configuration

Table	Purpose
core.status_definitions	Per-module status codes with color/badge
core.ui_themes	UI color palettes, one active per company
core.dashboard_metrics	KPI definitions with SQL, filter URL, allowed roles
core.dynamic_forms + core.form_fields	Dynamic form definitions with is_required flag
core.transaction_registry	Links modules to workflow definitions

3. Attendance Schema

Table	Key Columns
attendance.att_daily	id, employee_id, work_date, time_in, time_out, worked_hours, is_late, is_absent, late_minutes
attendance.att_shifts	id, company_id, code, name, start_time, end_time, days_of_week
attendance.att_shift_assignments	id, employee_id, shift_id, effective_from, effective_to
attendance.att_holiday_types	REGULAR, SPECIAL_NON_WORKING, etc. with pay multipliers
attendance.att_admin_overrides	Audit log of manual edits (employee_id, work_date, before, after, reason, by_user_id)
attendance.att_alerts	Generated alerts for late / absent patterns

4. Leave Management Schema

Table	Key Columns
leave_mgmt.lv_types	id, code, name, category, legal_basis, color, icon, is_paid, max_days_per_filing, gender_restriction
leave_mgmt.lv_balances	id, employee_id, leave_type_id, year, entitled, used, balance
leave_mgmt.lv_requests	id, employee_id, leave_type_id, date_from, date_to, days, reason, status
leave_mgmt.lv_holidays	id, company_id, hdate, name, htype, is_recurring, month_day
leave_mgmt.lv_policies	Per employment type: accrual_type, carry_over, monetization
leave_mgmt.lv_balance_adjustments	Non-destructive manual adjustments with audit trail

5. Payroll Schema

Table	Key Columns
payroll.pay_periods	id, company_id, period_code, date_from, date_to, payment_date, period_type
payroll.pay_runs	id, period_id, status (DRAFT/COMPUTING/COMPUTED/APPROVED/POSTED), run_type
payroll.pay_employee_payroll	id, run_id, employee_id, basic_pay, gross_pay, SSS/PhilHealth/HDMF/tax deductions, net_pay, allowances (PERA/RATA/ACA), government schemes (GSIS, coop)
payroll.pay_loans	id, employee_id, loan_type, principal, balance, amortization, start_date, end_date
payroll.pay_adjustments	Non-destructive adjustments (add-on or deduction) with reason + approver
payroll.v_payslip	VIEW joining employee payroll with run, period, department, position for printing

6. ESS / MSS Schema

Self-service and manager-service do not introduce new tables. They consume existing data via session.employee_id and apply privacy masking for own-record access.

7. Recruitment, PM, L&D, R&R Schemas

Schema	Representative Tables
rsp	requisitions, candidates, interviews, offers, csc_eligibilities, qualification_standards
pm	review_cycles, kpis, competencies, evaluations, calibrations, 360_feedback
ld	training_catalog, enrollments, certifications, training_sessions, attendance
rr	award_categories, awards, praise_wall, praise_config

8. DMS, Discipline, Health Schemas

Schema	Key Tables
dms	documents, document_categories (hierarchical), certificate_requests, certificate_types, retention_policies
discipline	cases, case_types, investigations, sanctions, hearings (with case_types keyed by gravity + legal_basis)
health	incidents, safety_reports, incident_types

9. Workforce Planning & Analytics

Schema / Feature	Tables
workforce_planning	scenarios (name, type, horizon_months, assumptions JSONB), headcount_projections, skill_demands
Analytics	Materialized views built on core/payroll/attendance; dashboard_metrics drives the admin dashboard
Report Builder	Per-user saved queries with role-scoped execution

10. Audit & Change Tracking

Every mutation on business-critical tables fires the `audit_logs.fn_capture_change()` trigger which writes a before/after snapshot into `audit_logs.change_log`. The trigger also captures `app.current_user_id` and `app.current_company_id` passed via SET LOCAL from the application.

11. Data Flow Examples

11.1 Filing a Leave Request

- 1 Employee submits POST `/leave/requests/new` (form data)

- 2 leave_mgmt.routes calls svc.create_request() which inserts into lv_requests with status=PENDING
- 3 Trigger writes audit_logs.change_log entry
- 4 Workflow engine picks up the record and creates inbox task for supervisor
- 5 Supervisor approves → status changes to APPROVED → trigger debits lv_balances.used
- 6 HR finalizes (optional) → status = COMPLETED

11.2 Running Payroll

- 1 Admin creates pay_period and pay_run (DRAFT)
- 2 Run triggers computation: for each employee, compute attendance, OT, leave days, allowances, then deductions (SSS, PhilHealth, HDMF, BIR tax) and net pay
- 3 Results written to pay_employee_payroll (run_id, employee_id)
- 4 Status flows: DRAFT → COMPUTING → COMPUTED → APPROVED → POSTED
- 5 Once POSTED, payslip_generated=true and employees see payslips in /me/payslips
- 6 Adjustments after posting go to pay_adjustments (non-destructive)

12. Data Retention

- Employee records: retained for the life of the company record (soft-delete via status)
- Payroll: retained 10 years per BIR requirements
- Attendance logs: retained 5 years
- Audit logs: retained 3 years (partitionable by month)
- Document retention: driven by dms.retention_policies per category